

MECH.5200 Numerical Methods for PDEs – Tentative Schedule Spring 2026

Table 1: Class and Assignments Tentative Schedule.

Item	Topics	Date
Class 1	Course Intro, Vectors, Matrices	21-Jan
Class 2	Finite differences / 1D FD implementa- tion / Error	28-Jan
HW #1 Assigned	Finite Differences Homework As- signed (1D, 2D, Mappings)	4-Feb
Class 3	2D Finite Differences (cont'd), Mappings in 1D, Implementation details	4-Feb
Class 4	2D Finite Differences w/ Mappings, incl. MATLAB Implementation	11-Feb
Project Proposal	Semester Group Project PRO- POSAL Due	13-Feb
Class 5	Classifying equations, FD for Parabolic Equations, Stability Analysis	18-Feb
Class 6	Hyperbolic Equations, FD for linear wave, Intro to Finite Volumes	25-Feb
Class 7	Finite Volumes	4-Mar
HW #1 Due	Homework Project #1 DUE	4-Mar
HW #2 Assigned	Hyperbolic / Conservation Equa- tions (FD and Finite Volumes)	4-Mar
BREAK	SPRING BREAK	11-Mar
Class 8	Finite Volume Methods 2D & Intro to FEM	18-Mar
HW #3 Assigned	Homework Project #3 Assigned	
Class 9	Finite Element Methods 1D, intro to 2D	25-Mar
HW #2 Due	Homework Project #2 DUE	29-Mar
Class 10	Finite Element Methods 2D	1-Apr
Class 11	Finite Element Methods 2D	8-Apr
Class 12	Discontinuous Galerkin FEM	15-Apr
HW #3 Due	Homework Project #3 DUE	15-Apr
HW #4 Assigned	Homework Project #4 Assigned	
Class 13	Boundary Element Method	22-Apr
Class 14	Project Presentations	29-Apr
HW #4 Due	Homework Project #4 DUE	29-Apr*

* If allowed, Homework #4 may be extended until exam week (Assigned Final Exam Day/Time).